# Cipla

*For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only OR for Specialist Use only*

**Ciprubicin**
**(Epirubicin Hydrochloride Injection 2 mg/ml)**

**COMPOSITION**
Each ml contains
Epirubicin Hydrochloride Ph. Eur………..2.0 mg
Sodium Chloride USP……………………9.0 mg
Water for Injection USP……………………q.s.

**DOSAGE FORM**
Intravenous or Intravesical Injection

**DESCRIPTION**
A clear, orange to red colour solution.

**PHARMACOLOGY**

**Pharmacodynamics**
Pharmacotherapeutic group: Antineoplastic agent. ATC code: L01D B03

The mechanism of action of Epirubicin is related to its ability to bind to DNA. Cell culture studies have shown rapid cell penetration, localisation in the nucleus and inhibition of nucleic acid synthesis and mitosis. Epirubicin has proved to be active on a wide spectrum of experimental tumours including L1210 and P388 leukaemias, sarcomas SA180 (solid and ascitic forms), B16 melanoma, mammary carcinoma, Lewis lung carcinoma and colon carcinoma 38. It has also shown activity against human tumours transplanted into athymic nude mice (melanoma, mammary, lung, prostatic and ovarian carcinomas).

**Pharmacokinetics**
In patients with normal hepatic and renal function, plasma levels after I.V. injection of 60-150mg/m2 of the drug follow a tri-exponential decreasing pattern with a very fast first phase and a slow terminal phase with a mean half-life of about 40 hours. These doses are within the limits of pharmacokinetic linearity both in terms of plasma clearance values and metabolic pathway. The major metabolites that have been identified are epirubicinol (13-OH-epirubicin) and glucuronides of epirubicin and epirubicinol.

The 4'-O-glucuronidation distinguishes epirubicin from doxorubicin and may account for the faster elimination of epirubicin and its reduced toxicity. Plasma levels of the main metabolite, the 13-OH derivative (epirubicinol) are consistently lower and virtually parallel those of the unchanged drug.

Epirubicin is eliminated mainly through the liver; high plasma clearance values (0.9 l/min) indicate that this slow elimination is due to extensive tissue distribution. Urinary excretion accounts for approximately 9-10% of the administered dose in 48 hours. Biliary excretion represents the major route of elimination, about 40% of the administered dose being recovered in the bile in 72 hours.

The drug does not cross the blood-brain-barrier. When Epirubicin is administered intravesically the systemic absorption is minimal.

**INDICATION**

Indicated for the treatment of: -breast cancer; -gastric cancer; -ovarian cancer; -small cell lung cancer; -lymphoma (non-Hodgkin's lymphoma); -advanced/metastatic soft tissue sarcoma; -superficial bladder cancer (Tis; Ta) In bladder cancer, it is also indicated in the prophylaxis of recurrence after transurethral resection of stage T1 papillary cancers and stage Ta multifocal papillary cancers (Grade 2 and 3).

**RECOMMENDED DOSE**

Epirubicin is intended for intravenous or intravesical administration only. It must not be administered by the intramuscular, subcutaneous or oral routes. Care in the intravenous administration of Epirubicin will reduce the chance of perivenous infiltration. It may also decrease the chance of local reactions, such as urticaria and erythematous streaking.

NOTE: The recommended lifetime cumulative dose limit is 1000 mg Epirubicin/m2 body surface area.

Under conditions of normal recovery from drug-induced toxicity (particularly bone marrow depression and stomatitis), the recommended dosage schedule in adults, as described below, is as a single intravenous injection administered at 21 day intervals. Standard doses are 75 to 90 mg/m2.

Epiorubicin produces predominantly haematological dose limiting toxicities which are predicted from the known dose-response profile of the drug. Based on the patient's haematological status the physician should determine the choice of dose. Higher doses, up to 135 mg/m2 as a single agent and 120 mg/m2 in combination, every 3-4 weeks have been effective in the treatment of breast cancer. In the adjuvant treatment of early breast cancer patients with positive lymph nodes, doses ranging from 100 mg/m2 to 120 mg/m2 every 3-4 weeks are recommended.

Careful monitoring in regards to increased myelosuppression, nausea, vomiting and mucositis are recommended in this high dose setting. Consideration should be given to the administration of lower starting doses (not exceeding 75-90 mg/m2) for heavily pre-treated patients, patients with pre-existing bone marrow depression or in the presence of neoplastic bone marrow infiltration. If Epirubicin

is used in combination with other cytotoxic drugs with potentially overlapping toxicities, the recommended dose per cycle should be reduced accordingly. While no specific dose recommendation can be made on the limited available data in patients with renal impairment, lower starting doses should be considered in patients with severe renal impairment (serum creatinine >5 mg/dL).

Intravesical Administration For the treatment of papillary transitional cell carcinoma of the bladder, a therapy of 8 weekly instillations of 50 mg is recommended. In the case of local toxicity (chemical cystitis) a dose reduction up to 30 mg is advised. For carcinoma in-situ, depending on the individual tolerability of the patient, the dose may be increased up to 80 mg. For prophylaxis of recurrences after transurethral resection of superficial tumours, 4 weekly administrations of 50 mg followed by 11 monthly instillations at the same dosage are recommended.

To avoid undue dilution with the urine, the patient should be instructed not to drink any fluid in the twelve hours prior to instillation. Intravesical administration is not suitable for the treatment of invasive tumours which have penetrated the muscular layer of the bladder wall.

Dose Modifications As clinical toxicity may be increased by the presence of impaired liver function, Epirubicin dosage must be reduced if hepatic function is impaired, according to the following table: Serum Bilirubin Levels Recommended Dose 20 - 50 µmol/L 1/2 normal dose Over 50 µmol/L 1/4 normal dose Haematological toxicity may require dose reduction, delay or suspension of Epirubicin therapy. Lower doses may be necessary if Epirubicin is used concurrently with other anti-neoplastic agents. Intravenous Administration It is recommended that Epirubicin be slowly administered into the tubing of a freely running intravenous infusion of Sodium Chloride Injection USP or 5% Glucose Injection USP.

The tubing should be attached to a Butterfly needle inserted preferably into a large vein. The rate of administration is dependent on the size of the vein and the dosage. However, the dosage should be administered in not less than 3 to 4 minutes. A direct push injection is not recommended due to the risk of extravasation, which may occur even in the presence of adequate blood return upon needle aspiration. Local erythematous streaking along the vein as well as facial flushing may be indicative of too rapid administration. A burning or stinging sensation may be indicative of perivenous infiltration and the infusion should be immediately terminated and restarted in another vein. Intravesical Administration Epirubicin, to be instilled using a catheter, should be retained intravesically for 1 hour.

The patient should be instructed to void at the end of this time. During instillation, the pelvis of the patient should be rotated to ensure extensive contact of the solution with the vesical mucosa.

Compatibility Epirubicin is compatible with the following infusion media:

- 0.9% Sodium Chloride
- 5% Glucose
- 0.9% Sodium Chloride with 5% Glucose

Epirubicin can be used in combination with other antitumour agents, but it is not recommended that it be mixed with these drugs in the same container. Heparin: Epirubicin should not be mixed with heparin as these drugs are incompatible. Until specific compatibility data are available, it is not recommended that Epirubicin be mixed with other drugs.

**CONTRAINDICATIONS**

Epirubicin is contraindicated in patients with marked myelosuppression induced by previous treatment with other antitumour agents or by radiotherapy and in patients already treated with maximal cumulative doses of other anthracyclines such as Doxorubicin or Daunorubicin. The drug is contraindicated in patients with current or previous history of cardiac impairment.

**WARNINGS AND PRECAUTIONS**

General

Epirubicin should be administered only under the supervision of qualified physicians experienced in antiblastic and cytotoxic therapy. Treatment with high dose Epirubicin in particular requires the availability of facilities for the care of possible clinical complications due to myelosuppression.

Initial treatment calls for a careful baseline monitoring of various laboratory parameters and cardiac function.

During each cycle of treatment with Epirubicin, patients must be carefully and frequently monitored. Red and white blood cells, neutrophils and platelet counts should be carefully assessed both before and during each cycle of therapy. Leukopenia and neutropenia are usually transient with conventional and high-dose schedules, reaching a nadir between the 10th and 14th day and returning to normal values by the 21st day; they are more severe with high dose schedules. Very few patients, even receiving high doses, experience thrombocytopenia (< 100,000 platelets/mm3).

Above cumulative dose of 900-1000 mg/m2 level the risk of irreversible congestive cardiac failure increases greatly. There is objective evidence that the cardiac toxicity may occur rarely below this range. However, cardiac function must be carefully monitored during treatment to minimise the risk of cardiac failure of the type described for other anthracyclines.

Heart failure can appear even several weeks after discontinuing treatment, and may prove unresponsive to specific medical treatment. The potential risk of cardiotoxicity may increase in patients who have received concomitant, or prior, radiotherapy to the mediastinal pericardial area.

In establishing the maximal cumulative doses of Epirubicin, any concomitant therapy with potentially cardiotoxic drugs should be taken into account.

It is recommended that an ECG before and after each treatment cycle should be carried out. Alterations in the ECG tracing, such as flattening or inversion of the T-wave, depression of the S-T segment, or the onset of arrhythmias, generally transient and reversible, need not necessarily be taken as indications to discontinue treatment.

Cardiomyopathy induced by anthracyclines, is associated with a persistent reduction of the QRS voltage, prolongation beyond normal limits of the systolic interval (PEP/LVET) and a reduction of the ejection fraction. Cardiac monitoring of patients receiving Epirubicin treatment is highly important and it is advisable to assess cardiac function by non-invasive techniques such as ECG, echocardiography and, if necessary, measurement of ejection fraction by radionuclide angiography.

**Renal Impairment**
Like other cytotoxic agents, Epirubicin may induce hyperuricaemia as a result of rapid lysis of neoplastic cells. Blood uric acid levels should therefore be carefully checked so that this phenomenon may be controlled pharmacologically. Epirubicin may impart a red colour to the urine for 1-2 days after administration.

**Hepatic impairment**
Before starting therapy and if possible during treatment, liver function should be evaluated (SGOT, SGPT, alkaline phosphatase, bilirubin). A cumulative dose of 900-1000 mg/m2 should only be exceeded with extreme caution with both conventional and high doses.

**DRUG INTERACTIONS**
It is not recommended that Epirubicin be mixed with other drugs. But Epirubicin can be used in combination with other anticancer drugs.

Cimetidine increases the formation of the active metabolite of epirubicin and the exposure of the unchanged epirubicin by pharmacokinetic interaction.

**UNDESIRABLE EFFECTS**
Apart from myelosuppression and cardiotoxicity, the following adverse reactions have been described:

Alopecia, normally reversible, appears in 60-90% of treated cases; it is accompanied by lack of beard growth in males.

Mucositis may appear 5-10 days after the start of treatment, and usually involves stomatitis with areas of painful erosions, mainly along the side of the tongue and the sublingual mucosa. Gastro-intestinal disturbances, such as nausea, vomiting and diarrhoea.

Hyperpyrexia.
Fever, chills and urticaria have been rarely reported; anaphylaxis may occur.

High doses of epirubicin have been safely administered in a large number of untreated patients having various solid tumours and have caused adverse events which are no different from those seen at conventional doses with the exception of reversible severe neutropenia (< 500 neutrophils/mm3 for < 7 days) which occurred in the majority of patients. Only a few patients required hospitalisation and supportive therapy for severe infectious complications at high doses.

During intravesical administration, as drug absorption is minimal, systemic side effects are rare; more frequently chemical cystitis, sometimes haemorrhagic, has been observed.

Haematological
The occurrence of secondary acute myeloid leukaemia with or without a pre-leukaemic phase has been reported rarely in patients concurrently treated with Epirubicin in association with DNA- damaging antineoplastic agents. Such cases could have a short (1-3 year) latency period.

**Pregnancy and Lactation**

**Pregnancy**
There is no conclusive information as to whether epirubicin may adversely affect human fertility or cause teratogenesis. Experimental data, however, suggest that epirubicin may harm the foetus. This product should not normally be administered to patients who are pregnant. Like most other anti-cancer agents, epirubicin has shown mutagenic and carcinogenic properties in animals.

**Lactation**
This product should not normally be administered to mothers who are breast-feeding.

**OVERDOSAGE**
Very high single doses of epirubicin may be expected to cause acute myocardial degeneration within 24 hours and severe myelosuppression within 10-14 days. Treatment should aim to support the patient during this period and should utilise such measures as blood transfusion and reverse barrier nursing. Delayed cardiac failure has been seen with the anthracyclines up to 6 months after the overdose. Patients should be observed carefully and should, if signs of cardiac failure arise, be treated along conventional lines.

**SHELF LIFE**
24 months

**STORAGE AND HANDLING INSTRUCTIONS**
Store in refrigerator, $2^o$ – $8^oC$ ($36^o$ to $46^oF$). Do not freeze, Protect from light. Retain in carton until time of use.

**PACK SIZE**

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| **Epirubicin Hydrochloride Injection 50 mg/25 ml** | Vial of 30 ml having fill value of 25 ml Injection |

**REGISTRATION HOLDER IN MALAYSIA**
CIPLA MALAYSIA SDN BHD
Suite 1101, Amcorp Tower,
Amcorp Trade Centre,
18 Persiaran Barat,
46 050 Petaling Jaya,
Selangor, Malaysia

**Mfd. by**
**CIPLA LTD.,**
S-103 to S-105,
S-107 to S-112,
L-147 to L-147-1
Verna Industrial Estate
Goa – 403722. India

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